

GUIDE-LLM: A reporting checklist for large language models in behavioural science

The GUIDE-LLM checklist provides a standardized framework for reporting studies that use large language models (LLMs) in the behavioral and social sciences. It aims to promote transparency, reproducibility, and ethical accountability across all stages of LLM-based research.

How to complete the checklist:

Fill in each item with brief, specific information about how LLMs were used in your study. Where an item does not apply, write “N/A” and, if helpful, note why. If multiple LLMs were used for different purposes, complete the relevant sections separately for each model. You may refer to sections or appendices in your manuscript rather than repeating text.

Scope of LLM use	Answer
Item A.1: LLMs were used in this project for: Explanation: Briefly describe how and for what purposes LLMs were used in the study. This may include one or multiple stages of the research workflow, depending on the project’s design and aims. The following examples illustrate common use cases: • Research design (e.g., hypothesis generation, literature search, or creating surveys/stimuli). • Data processing (e.g., transcription, translation, data extraction, or data cleaning). • Analysis (e.g., data labeling, summarization, pattern detection, statistical analysis, or code generation). • LLM as research object (e.g., studying LLM behavior, benchmarking LLMs, or bias assessment of LLMs). • LLM as an evaluator (e.g., evaluating model output, assessing the quality of human labeling/annotation) • Participant-facing settings (e.g., LLM used as an intervention, studying human interactions with LLM chatbots). • Communication (e.g., paper writing, editing, or reviewing). Depending on the specific use case described here, different checklist items may later be relevant, and, in many cases, it may be necessary that later items in the checklist are reported separately for each use case.	LLM as research object. LLMs were used to identify and develop effective LLM detection tests embedded in an online survey.
Item A.2: Degree of automation (human-in-the-loop vs. fully automated): Explanation: Indicate how much human oversight was involved. Specify whether each output was reviewed, edited, or approved by a person, or whether outputs were used automatically without supervision. For participant-facing tasks, state whether humans checked outputs before showing them to participants or whether participants interacted with the LLM directly. Specify who provided oversight (e.g., student assistant, expert, PI).	Both human-in-the-loop and fully automated procedures were used in this project. See Supplemental Information (Method, AI Agents section) for further detail.

Model/system details	Answer
Item B.1: Model name, including provider, model size, exact version/ID, date of access, and source link (if possible): Explanation: Report the exact model names (including provider, version, and date accessed). Avoid generic labels like “ChatGPT” or “GPT-4”; instead, use detailed model names such as “GPT-4o-mini-2024-12-17 (OpenAI)” or “Llama-3.1-8B (Meta; accessed via HuggingFace in May 2025)”. For locally deployable models, please also enter a source link (e.g., the URL to the HuggingFace page). If multiple models were tested, it is encouraged to name them and briefly explain which one was used in the final study and why. When multiple models served different purposes, specify their respective roles, consistent with your response to Item A.1.	Three models were used in browser-agent mode: ChatGPT Atlas, ManusAI, and Perplexity Comet. In addition, one agent harness, browser-use v0.13.1 operated by Claude Opus 4.8 through the API was used. See Supplemental Information (Method, AI Agents section) for full description of technical details and date of access.
Item B.2: Model access (e.g., API, web interface, local) and context mode (e.g., chat mode or separate calls): Explanation: Note how you accessed the models (e.g., API, web interface, local installation) and whether you used LLMs in chat mode (ongoing conversation) or stateless mode (separate prompts). Mention the exact API name and version, since different access modes may influence responses (e.g., due to differences in model routing).	Models were accessed via chat in browser-agent mode.
Item B.3: Relevant LLM configurations reported (as applicable), such as temperature, max tokens, seed, and number of runs: Explanation: List any configuration settings that may affect outputs, such as: • temperature (which controls randomness of the model’s output) • Sampling parameters such as top_k, top_p, max tokens (which limit the candidate token set or enforce length constraints) • Penalties that discourage repetition (e.g., a frequency penalty to reduce the likelihood of tokens proportional to how often they have already appeared; a presence penalty reduces the likelihood of any token that has appeared at least once) • Stop sequences (which halt generation when such a specified sequence is produced; e.g., ["\n\n", "END"]) • Number of completions or runs (which is often used to capture variability in outputs across repeated generations) • Quantization level (e.g., FP16, INT8, INT4) to change numerical precision beyond the default • Reasoning-related settings, such as whether a specific structured reasoning was enabled, the specified reasoning effort level (e.g., low/medium/high or numerical settings that influence the depth of the reasoning), and any compute or inference budget constraints tied to the chosen reasoning mode	The Browser-use bot where Claude Opus 4.8 was run with extra high reasoning effort. No other custom configurations were set except.

Item B.4: Customization: Explanation: Check and describe any modifications or extended capabilities incorporated into your LLM setup beyond standard inference. This includes, but is not limited to: • Fine-tuning (e.g., via LoRA; Low-Rank Adaptation) used to adapt a pretrained model to domain-specific data. • Retrieval-augmentation generation (RAG), where the model retrieves relevant information from external sources (e.g., databases or document collections) during inference. • Automated prompt optimization (e.g., DSPy) that treat prompts as trainable parameters. • Web search integration, indicating whether the LLM was able to access and retrieve information from the Internet. • Agentic workflows, including multi-step reasoning processes or delegated actions such as tool/function calling (e.g., via LangChain, AutoGPT, CrewAI). • Post-training refinements, including alignment or optimization techniques used to adjust model behavior after pretraining (e.g., reinforcement learning from human feedback (RLHF), direct preference optimization (DPO)). The goal is to specify any added customizations or provider-specific features that meaningfully shape system behavior in order to enable others to understand and accurately reproduce your setup.	X Base model ☐ Fine-tuning ☐ RAG (retrieval-augmented) ☐ Automated prompt optimization ☐ Tool/function calling ☐ Web search ☐ Agentic workflows ☐ Other adaptations (e.g., safety mechanisms) Description: N/A.
Item B.5: Did the LLM session(s) include persistent memory across interactions? Explanation: Indicate whether the LLM could “remember” previous conversations (i.e., had persistent memory). Unless such memory is disabled, there may also be spillover effects from other chat windows or prior conversations, which can influence outputs even when not intended.	☐ Yes X No ☐ N/A

Prompts	Answer
Item C.1: Exact prompt(s) reported: Explanation: Whenever possible, include the exact text of prompts you used, including in-context examples or demonstrations provided to the LLM. Even small wording changes, formatting, or ordering of examples can substantially affect outputs. If full prompts cannot be shared (e.g., due to privacy or length), include a redacted or representative example or link to the full prompt in a repository (e.g., OSF, GitHub). If prompts were iteratively refined during the study, document the nature and extent of these refinements, including whether changes were pre-specified or made in response to interim outputs, to ensure transparency regarding potential optimization or selective reporting.	For the full list of prompts used, see Supplemental Information (Materials, AI Agent Prompts section).
Item C.2: System-wide instructions (if any): Explanation: Note any <i>system-level</i> instructions that guide the model’s general behavior (e.g., “You are a helpful assistant.”). These are commonly not directly visible but can be accessed through the API.	N/A.

Data inputs & privacy	Answer
Item D.1: Handling of personal or sensitive data (if any) (e.g., consent for data processing): Explanation: If any personal, sensitive, or identifiable data were processed, describe how they were handled in compliance with ethical standards and data protection laws. Researchers should indicate whether participants explicitly consented to their data being analyzed with an LLM, particularly when proprietary, cloud-based models are used. Such processing typically involves transferring data to a private company that may retain them indefinitely, which raises additional ethical and legal considerations. Beyond consent, describe how sensitive or identifiable data were handled (e.g., de-identification, anonymization, masking) and whether the LLM provider offers safeguards such as excluding inputs from training or storage. Clarify where data were stored or processed and how applicable legal/ethical requirements were met. If relevant, address cross-border transfers, as data may be stored in jurisdictions with different privacy laws (e.g., EU vs. US), with implications for compliance with GDPR, HIPAA, or other frameworks. For context, some providers (e.g., OpenAI) may log or inspect prompts even when the data are not used for model training. For sensitive datasets, zero-retention configurations may be required (e.g., the MIMIC datasets can only be used with OpenAI models if a zero-retention checkpoint is enabled).	N/A.

Validation & interpretation	Answer
Item E.1: Human validation of LLM outputs: Explanation: If relevant, describe whether and how human reviewers examined the model's outputs, and the degree of independence they had in doing so. Specify the reviewers' roles (e.g., domain experts, research assistants, subject-matter specialists) and relevant expertise, as well as how many reviewers participated and how their work was organized. Indicate whether outputs were independently annotated, double-checked by multiple reviewers, or merely approved or edited post-hoc by a lead author or investigator. Clarify what dimensions of performance were examined. These may include known performance metrics from ML/AI such as accuracy or other metrics like citation correctness, hallucination detection, agreement or inter-rater reliability. State whether qualitative judgments, quantitative metrics, or both were used. If outcome assessment required subjective interpretation, describe assessor qualifications, instructions provided, and relevant demographics. Describe the selection procedure for the reviewed outputs—whether all outputs were examined, a random sample was drawn, or specific cases (e.g., rare events or high-stakes responses) were oversampled to capture potential rare or critical errors. Further report how reviewers were trained or instructed, what criteria or rating scales they used, and how disagreements were resolved. For multi-reviewer settings, provide any inter-rater or inter-assessor reliability statistics (e.g., Cohen's κ or Krippendorff's α). Finally, note whether reviewer feedback was used purely for validation or also to refine prompts, retrain models, or adjust study procedures.	X Yes ☐ No ☐ N/A Description: Humans were involved in assessing the performance of the LLM in its ability to complete the task without requesting the user's input and its ability to pass different LLM detection tests. To monitor the first, three humans (two PhD students and one RA) tracked the number of user-control requests for each trial and, during prompt development, tweaked the prompt to reduce requests where necessary. For monitoring performance on the detection tests, two PhD students analyzed test fail rates and, during prompt development, adjusted the prompt to improve the pass rate.
Item E.2: Describe any relevant post-processing (e.g., filtering in case of format mismatches, unit conversions, etc.): Explanation: Describe any steps you took to clean or reformat LLM outputs (e.g., converting "positive/neutral/negative" to numeric codes, handling missing values, removing malformed entries). State how you handled inconsistent or unusable outputs and whether corrections were made with an automated script or manually. For example, when generating quantitative estimates (e.g., word counts, probabilities, or durations), the model may return values embedded in free text (e.g., "3.5 seconds") that require parsing and conversion into standardized numerical units. Post-processing steps should be described clearly, including how formatting errors, null responses, or inconsistent output structures were handled, whether automated scripts or manual corrections were used, and whether any data were excluded or reinterpreted as a result. Report how outputs were selected, extracted, or consolidated when multiple generations were produced.	N/A

Reproducibility	Answer
Item F.1: Code/notebooks/scripts for LLM calls shared: Explanation: Indicate whether you have shared materials such as code, prompts, logs, or transcripts. Make sure sensitive information (e.g., API keys, private data) is removed. For code, make sure to add a README file.	☐ Yes ☐ No X N/A Link/DOI:

Competing interests	Answer
Item G.1: Funding, support, or other relevant relationships (including in-kind access to compute or models, or professional affiliations): Explanation: Disclose any current or past funding, support, or other relevant relationships with entities that have a financial interest in LLMs (this includes not just AI companies like OpenAI, Anthropic, but also tech companies developing or investing in AI, e.g., Google, Meta, Microsoft). This could include (but is not limited to): research funding from or collaborative research with a company with an interest in LLMs for this project or any other project within the past years; in-kind access to compute or models; current or former professional affiliations with a company with an interest in LLMs; personal investments (e.g., stocks) in companies with an interest in LLMs; familial relationship with an employee of a company with an interest in LLMs; etc. Disclose these relationships regardless of whether or not you believe they impacted the research.	☐ Yes. Description: X No

Optional items	Answer
Justification for the specific LLM choice along the following dimensions: Explanation: Briefly explain why you chose this specific model for your study and thereby explain why the selected model is suited to the research needs. Consider: ● Performance: Explain why this model's accuracy or size was appropriate for the research task, and, if relevant, consider compute cost, scalability, and real-time feasibility. ● Transparency: Was the model open-weight or closed-weight? Do we know what training data the model was exposed to or what the training objective was? If not, what are the implications for interpreting results? In terms of openness, different levels can be distinguished, in which only the model weights are publicly available ("open-weight") versus LLMs where the entire training process is available to allow for further customization and transparency of the LLM itself ("open-source"). ● Reproducibility: Can the model be shared (so results can be reproduced in the future), or is access limited to proprietary/cloud services (so the access to the model may be altered, restricted, or discontinued)? ● Ethical considerations: Are there ethical reasons (including safety) for the model choice, such as running a model locally to avoid sharing sensitive data? ● Other determinants can be the cost or ease-of-use (e.g., due to accessibility of certain models via university infrastructures), if their benefits outweigh trade-offs in other dimensions.	Performance – Description: Models were selected for their accessibility to a typical online survey platform user: off-the-shelf, easily accessible models requiring minimal technical sophistication. Transparency – Description: All models used in this project were proprietary, closed-weight models. However, the browser-use harness was open-source. Reproducibility – Description: Access is limited to proprietary services. Ethical considerations – Description: N/A. Others (e.g., cost, ease-of-use): – Description: Ease-of-use is an important determinant as users of online survey platforms (our target demographic for this LLM use-case) are unlikely to have the technical sophistication to build a custom LLM, so widely available models that work out of the box were prioritized. To examine whether there may be some guardrails on the commercial agents that prevented it from passing some of our tests, we additionally ran our enhanced prompt on an open-source agent framework.
Discussion of the rationale for the prompt design: Explanation: Explain how you designed your prompts. For example, indicate whether you used a structured format (e.g., explicit task description, definitions, step-by-step instructions, and output constraints), followed established prompt engineering guidelines or prior literature, adapted prompts from earlier studies, or relied on automated prompt optimization tools. Clarify whether the design was iterative (e.g., refined through pilot testing or error analysis), whether few-shot examples were included and how they were selected, and whether prompts were standardized across models to ensure comparability.	X Yes. Description: See Supplemental Information (Method, AI Agents section) for discussion of prompt design. ☐ No ☐ N/A

Comparison against other methods/LLMs: Explanation: If you compared different models, prompts, or methods, describe how you made the comparison and what criteria you used (e.g., accuracy). State whether you used the same inputs across methods and how you ensured fair comparison. If you considered but excluded alternative methods/LLMs, briefly explain why. If such comparisons are made, report the exact benchmarking procedure and the level of agreement between methods. If other LLMs or methods were considered but not used, briefly explain the rationale (e.g., cost, access, performance, or ethical considerations).	X Yes. Description: Three LLMs were tested and compared using the same criteria: evading detection across multiple LLM detection tests. The open-source agent was used to validate the results of the commercial agents with an open-source agent framework that was unlikely to have the guardrails that the commercial agents potentially have. ☐ No ☐ N/A
Training data leakage risks addressed: Explanation: Note whether you considered the chance that your evaluation materials (e.g., test data, questionnaires) were already part of a model's training data. If relevant, describe steps taken to mitigate or detect leakage, such as novelty checks, re-phrasing or re-structuring items, or using hold-out materials not widely available.	☐ Yes ☐ No X N/A
Potential risk of bias or other systematic differences in LLM behavior that could affect the study's conclusions? Explanation: Reflect on whether the LLM might perform differently across groups, languages, or contexts in ways that could affect your conclusions. If you checked for or mitigated such bias, describe how (e.g., subgroup analyses, balanced sampling, sensitivity checks). Where applicable, researchers may consult dedicated bias and fairness checklists for structured guidance on identifying and addressing these issues.	X Yes. Description: More sophisticated or tailored models may be able to pass the LLM detection tests we found to be effective. ☐ No
Conversation transcripts: Explanation: For studies where participants or researchers interacted directly with an LLM (e.g., chat-based interventions, behavioral experiments, or coding tasks), provide anonymized transcripts or representative examples of the interactions. Redact any personally identifiable or sensitive information.	☐ Yes, shared without sensitive information. Location: ☐ No. Reason for not sharing: X N/A
Discuss relevant ethical implications of the research: Explanation: Briefly discuss the broader ethical considerations of your study beyond procedural compliance (e.g., IRB approval). Address potential risks and safeguards, including: • Participant well-being: foreseeable harms (emotional, reputational, financial, environmental), inclusion/exclusion criteria, effects on vulnerable groups, accessibility, use of deception and debriefing, autonomy and informed consent (including clear disclosure of AI involvement). • Fairness and equity: whether outputs or outcomes may reflect or amplify bias or discrimination; note any subgroup analyses or mitigation strategies. • Safety: broader risks (such as from agentic or autonomous behavior), or safeguards for harmful or adverse outputs (e.g., moderation layers, content filters, human-in-the-loop review, escalation protocols, red-teaming, dual use).	N/A.
Computational resources (e.g., API call counts, tokens, financial costs, or compute time): Explanation: Report the computational resources required for the study. This may include API call counts, total tokens processed, financial costs, runtime, number and type of GPUs/CPU's used, and any other relevant compute metrics.	N/A.

How to cite:

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